

Iniyan Andrews Joseph

M.Sc. Computer Science (Data Science)

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🌐 [LinkedIn](#)

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| Calicut, India

RESEARCH INTERESTS & PHD ALIGNMENT

Core Interests: Ensemble Feature Selection, Dimensionality Reduction, Supervised Learning (Classification), Evolutionary Algorithms (GA, PSO), Financial Time-Series Forecasting, and Stock Market Prediction.

EDUCATION

NIELIT, Calicut

Calicut, India

M.Sc. in Computer Science (Specialization in Data Science)

Expected Completion: May 2026

- **Current CGPA:** 8.55/10.0
- **Relevant Coursework:** Mathematics for Machine Learning, Advanced Statistical Techniques, Probability & Statistics, Data Analytics & Visualization, Introduction to Data Science & AI.

The American College, Madurai

Madurai, India

B.Sc. in Computer Science

2021 – 2024

- **CGPA:** 7.4/10.0
- **Key Projects:** Numerical & Statistical Methods, Database Management Systems.

RESEARCH & TECHNICAL PROJECTS

Short-Term Stock Prediction: Supervised Learning (Project HOPE)

[View on GitHub](#) 🐙

Independent Researcher

Python, Scikit-Learn, LSTM, TensorFlow

- Conducted rigorous statistical analysis to identify **S&P 500** and Nifty 50 stocks primed for 1% daily gains, framing the problem as a supervised learning classification task.
- Built and evaluated robust feature engineering pipelines to process high-noise financial data, isolating predictive signals and discarding redundant features to prevent deep learning model collapse.
- Demonstrated that variance-reducing, regularized linear models (190.65% Return, 2 Sharpe) significantly outperformed complex LSTM models by avoiding overfitting on high-dimensional financial datasets.

High-Frequency Trading using Deep Machine Learning

[View on GitHub](#) 🐙

Lead Researcher

Python, PyTorch, OpenAI Gym

- Engineered an autonomous trading agent that processed a highly complex **42-dimensional feature vector** representing dynamic market states.
- Addressed the curse of dimensionality by validating feature importance; utilized out-of-sample stress testing and look-ahead bias checks to ensure generalized, robust performance.
- Achieved a **Sharpe Ratio of 1.4** by optimizing statistical metrics across both bull and bear market regimes.

Jaguu: Multimodal Generative AI Assistant

[View on GitHub](#) 🐙

Lead Developer

Python, Llama3, Ollama, Computer Vision

- Built a privacy-first, local AI integrating Llama3 for core reasoning and custom Speech-to-Text pipelines, emphasizing optimized, low-latency Python code execution and algorithmic efficiency.

TECHNICAL SKILLS

- **Machine Learning:** Supervised Learning (Classification/Regression), Feature Selection, Dimensionality Reduction, Ensemble Learning, Evolutionary Algorithms (GA, PSO), PyTorch, TensorFlow.

- **Data Science & Stats:** Scikit-Learn, Pandas, NumPy, Matplotlib, Statistical Analysis, Graph Clustering.
- **Financial Analytics:** Stock Market Prediction (S&P 500), Time-Series Analysis, Risk Management Metrics (Sharpe, Sortino, Drawdown), Dealing with Concept Drift.
- **Programming:** Python (Advanced), C++, SQL, R.
- **Tools:** Git/GitHub, Docker, Linux, VS Code, LaTeX, Jupyter.

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ACADEMIC REFEREES

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